

Digital Image Processing

Lecture # 10 **Color Processing**

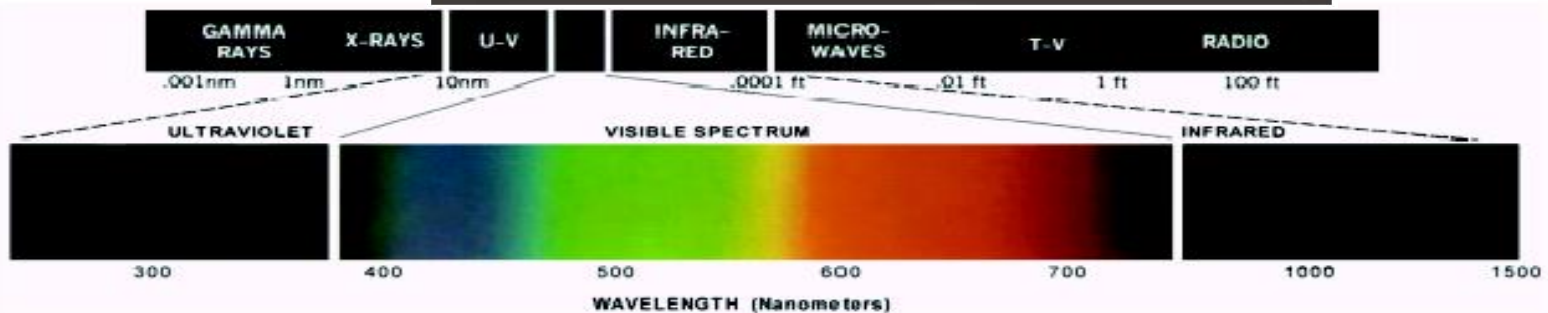
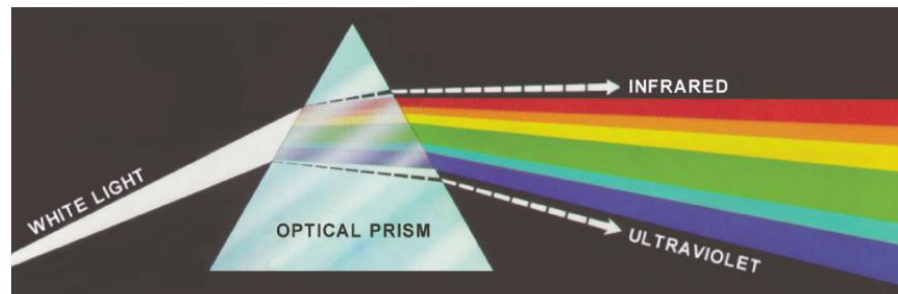
COLOR IMAGE PROCESSING

COLOR IMAGE PROCESSING

- Color Importance
 - Color is an excellent descriptor
 - Suitable for object Identification and Extraction
 - Discrimination
 - Humans can distinguish thousands of color shades and intensities but few shades of gray levels
- Color Image Processing
 - Full-Color Processing
 - Color is acquired with a full-color sensor
 - Pseudo-Color Processing
 - Assigning colors to monochrome images

COLOR FUNDAMENTALS

- Colors that humans perceive in an object are determined by the nature of the light reflected from the object
- Visible light is composed of a relatively narrow band of frequencies in the electromagnetic spectrum
- A body that reflects light that is balanced in all visible wavelengths appears white to the observer
- A body that favours reflectance in a limited range of the visible spectrum exhibits some shades of color
- Green objects reflect light with wavelengths primarily in the 500 to 570 nm range while absorbing most of the energy at other wavelengths



HUMAN PERCEPTION OF COLOR

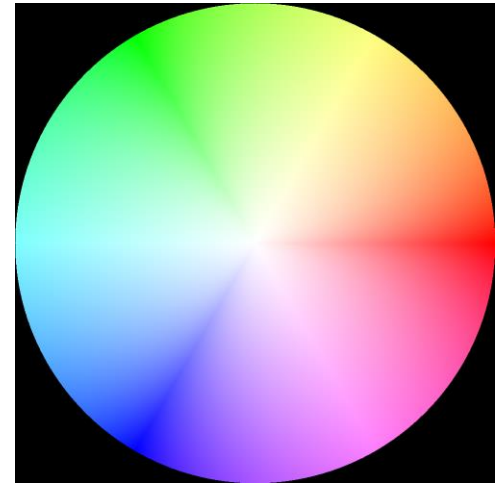
- Retina contains receptors
 - Cones
 - Day vision, can perceive color tone
 - Red, green, and blue cones
 - Rods
 - Night vision, perceive brightness only
- Color sensation
 - Luminance (brightness)
 - Chrominance
 - Hue (color tone)
 - Saturation (color purity)

Monochromatic images

- Image processing - static images -
- Monochromatic static image - continuous image function $f(x,y)$
 - arguments - two co-ordinates (x,y)
- Digital image functions - represented by matrices
 - co-ordinates = integer numbers
 - Cartesian (horizontal x axis, vertical y axis)
 - OR (row, column) matrices
- Monochromatic image function range
 - lowest value - black
 - highest value - white
- Limited brightness values = gray levels

Chromatic images

- Colour
 - Represented by vector not scalar
 - Red, Green, Blue (RGB)
 - Hue, Saturation, Value (HSV)
 - luminance, chrominance (Yuv , Luv)



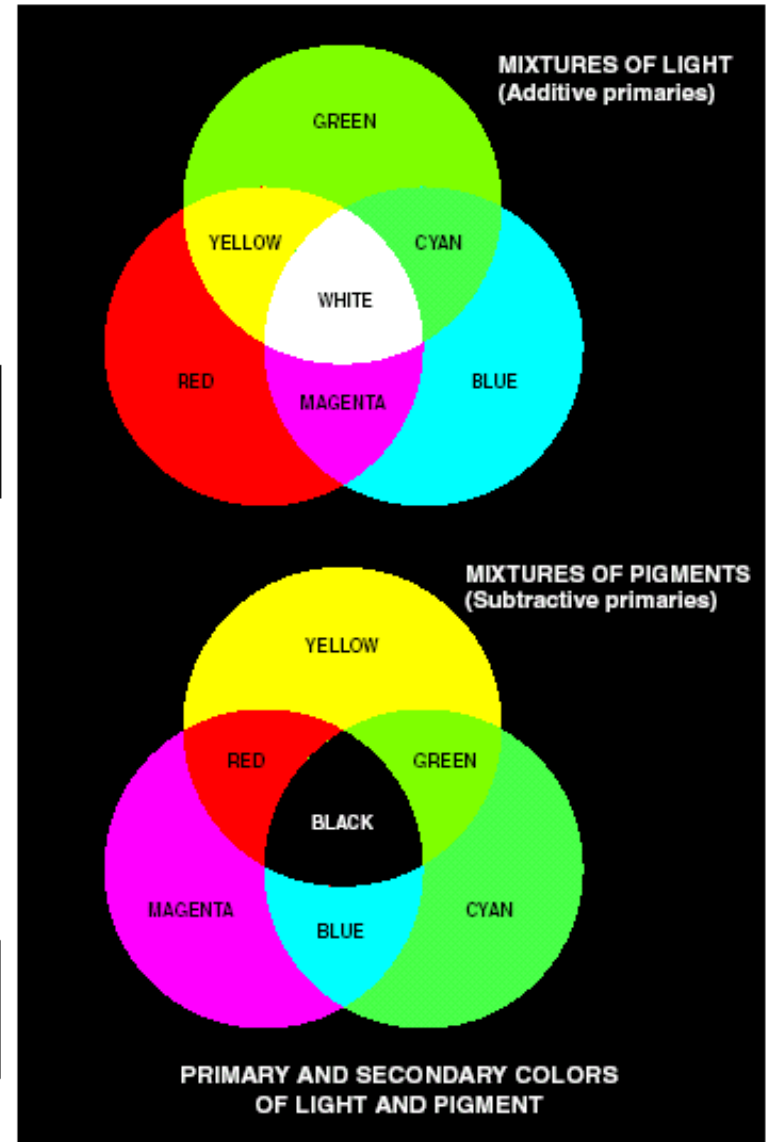
PRIMARY AND SECONDARY COLORS OF LIGHT AND PIGMENTS

- The primary colors can be added to produce the secondary colors of light
- The primary colors of light and primary colors of pigments are different

Magenta = Red + Blue
Cyan = Blue + Green
Yellow = Green + Red

- For pigments, a primary color is defined as one that absorbs a primary color of light and reflects the other two
- Therefore, the primary colors of pigments are magenta, cyan, and yellow

Magenta = White - Green
Cyan = White - Red
Yellow = White - Blue



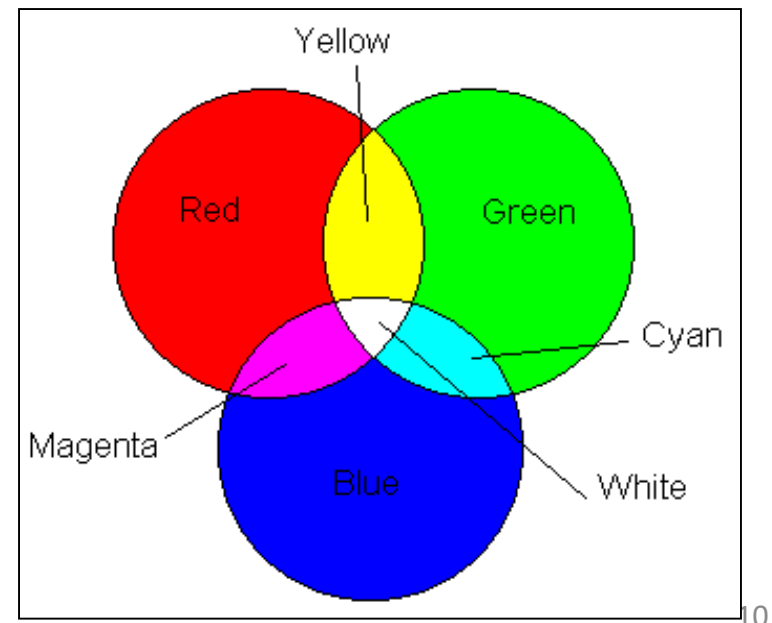
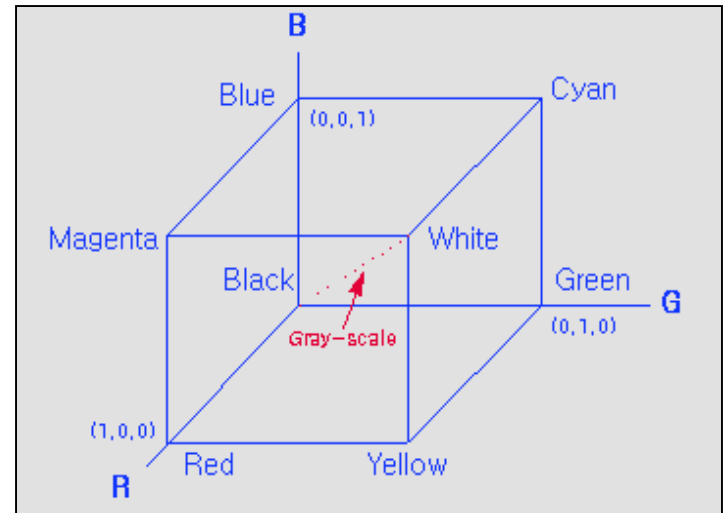
COLOR MODELS

- Color Model
 - Specify colors in a standard way
 - A coordinate system that each color is represented by a single point.
- Most used models:
 - RGB model (Monitor/TV)
 - CMY model (3-color Printers)
 - HSI model (Color Image Processing and Description)

RGB COLOR MODEL

- Pixel Depth: The number of bits used to represent each pixel in RGB space.
- Full-color image: 24-bit RGB color image.
 - (R, G, B) = (8 bits, 8 bits, 8 bits)
 - Number of colors:

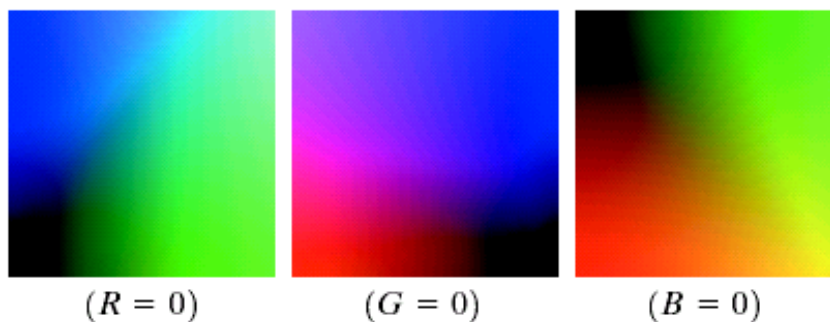
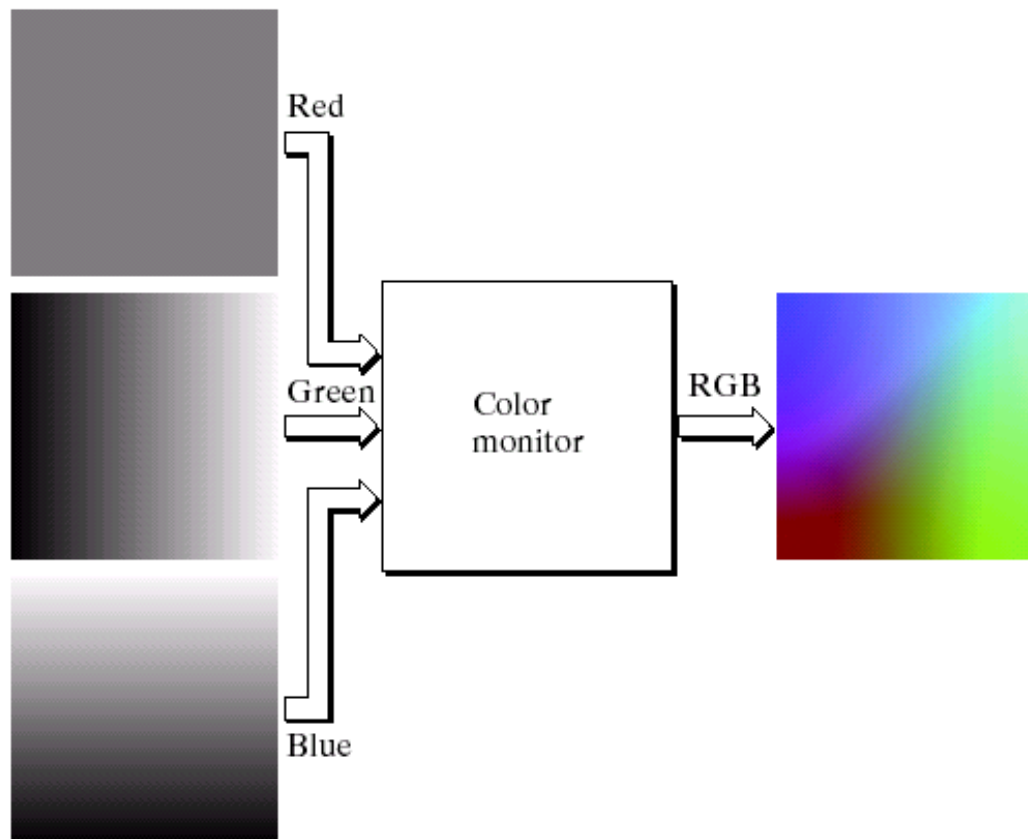
$$(2^8)^3 = 16,777,216$$



a
b

FIGURE 6.9

(a) Generating the RGB image of the cross-sectional color plane $(127, G, B)$.
(b) The three hidden surface planes in the color cube of Fig. 6.8.



COLOR IMAGE - RGB

Color Image



a b
c d

FIGURE 6.38

(a) RGB image.
(b) Red
component image.
(c) Green
component.
(d) Blue
component.

R-Channel

G-Channel



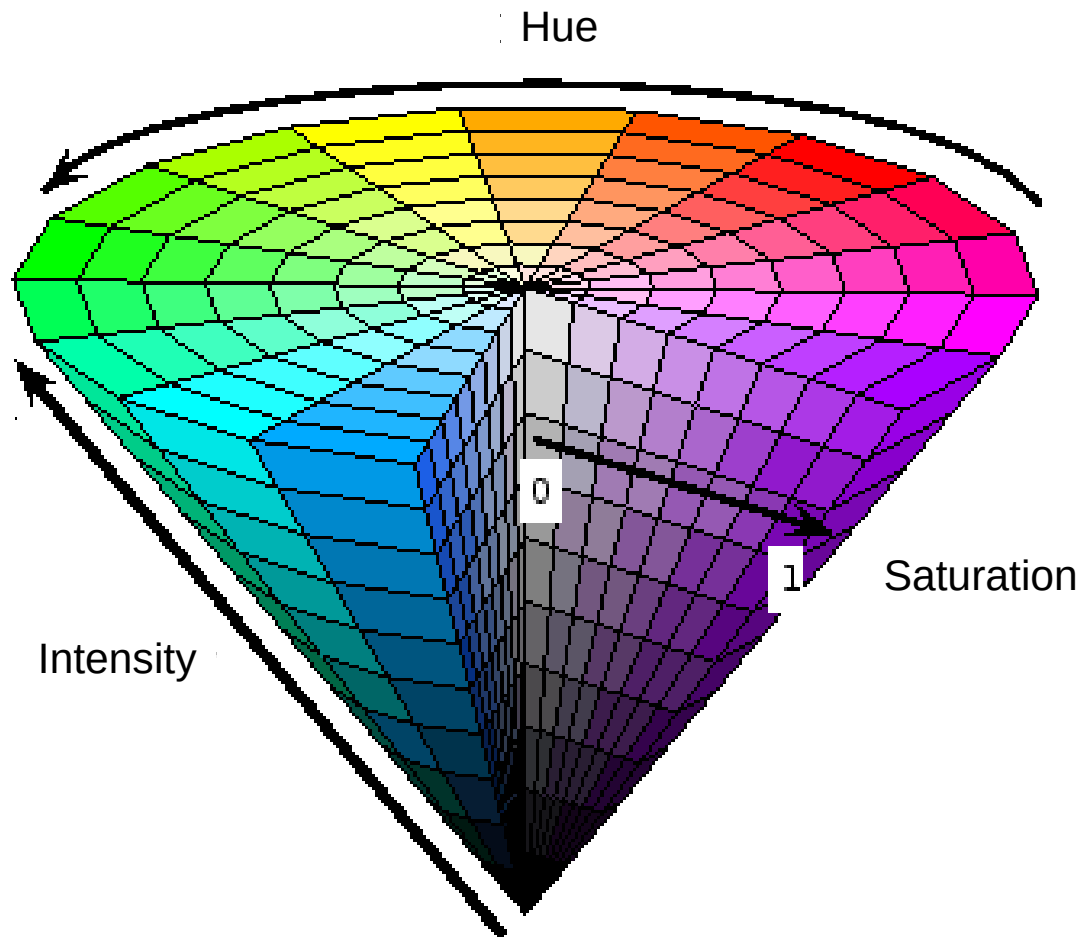
B-Channel

HSI COLOR MODEL

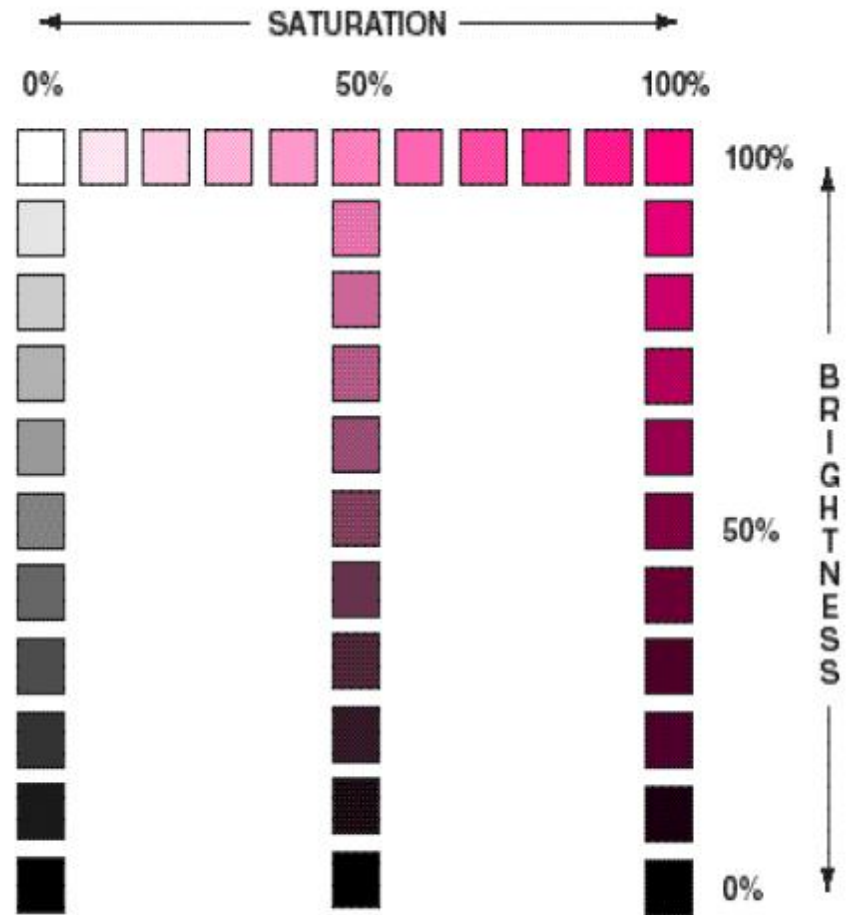
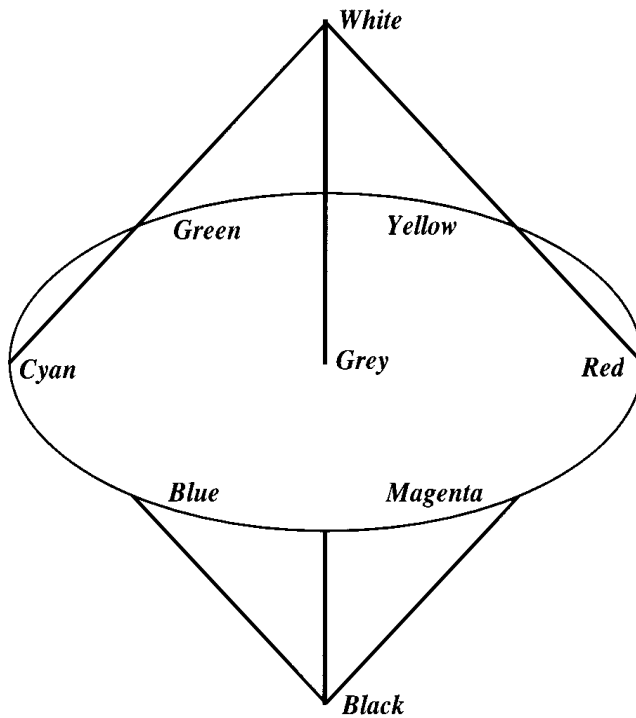
- Human description of color is Hue, Saturation and Brightness:
- Hue
 - represents dominant color as perceived by an observer. It is an attribute associated with the dominant wavelength.
- Saturation
 - refers to the relative purity or the amount of white light mixed with a hue. The pure spectrum colors are fully saturated.
 - Pure colors are fully saturated.
 - Pink is less saturated.
- Intensity
 - reflects the brightness.



HSI Color Model



HSI COLOR MODEL- SINGLE HUE



HSI Color Model

- **Hue** is defined as an angle
 - 0 degrees is **RED**
 - 120 degrees is **GREEN**
 - 240 degrees is **BLUE**
- **Saturation** is defined as the percentage of distance from the center of the HSI triangle to the pyramid surface.
 - Values range from 0 to 1.
- **Intensity** is denoted as the distance “up” the axis from black.
 - Values range from 0 to 1

Converting from RGB to HSI

$$\theta = \cos^{-1} \left\{ \frac{\frac{1}{2} [(R - G) + (R - B)]}{[(R - G)^2 + (R - B)(G - B)]^{\frac{1}{2}}} \right\}$$

$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360 - \theta & \text{if } B > G \end{cases}$$

$$S = 1 - \frac{3}{(R + G + B)} [\min(R, G, B)]$$

$$I = \frac{1}{3} (R + G + B)$$

Conversion Between RGB and HSI

- Converting color from RGB to HSI

$$H = \begin{cases} \theta & \text{if } B \leq G \\ 360 - \theta & \text{if } B > G \end{cases} \quad \text{with } \theta = \cos^{-1} \left\{ \frac{\frac{1}{2}[(R-G) + (R-B)]}{\left[\frac{1}{4}[(R-G)^2 + (R-B)(G-B)] \right]^{\frac{1}{2}}} \right\}$$
$$S = 1 - \frac{3}{(R+G+B)}[\min(R, G, B)]$$
$$I = \frac{1}{3}[R+G+B]$$

- Converting color from HSI to RGB

RG sector ($0 \leq H < 120$)

$$B = I(1 - S)$$

$$R = I \left[1 + \frac{S \cos H}{\cos(60 - H)} \right]$$

$$G = 1 - (R + B)$$

GB sector ($120 \leq H < 240$)

$$R = I(1 - S)$$

$$G = I \left[1 + \frac{S \cos(H - 120)}{\cos(60 - (H - 120))} \right]$$

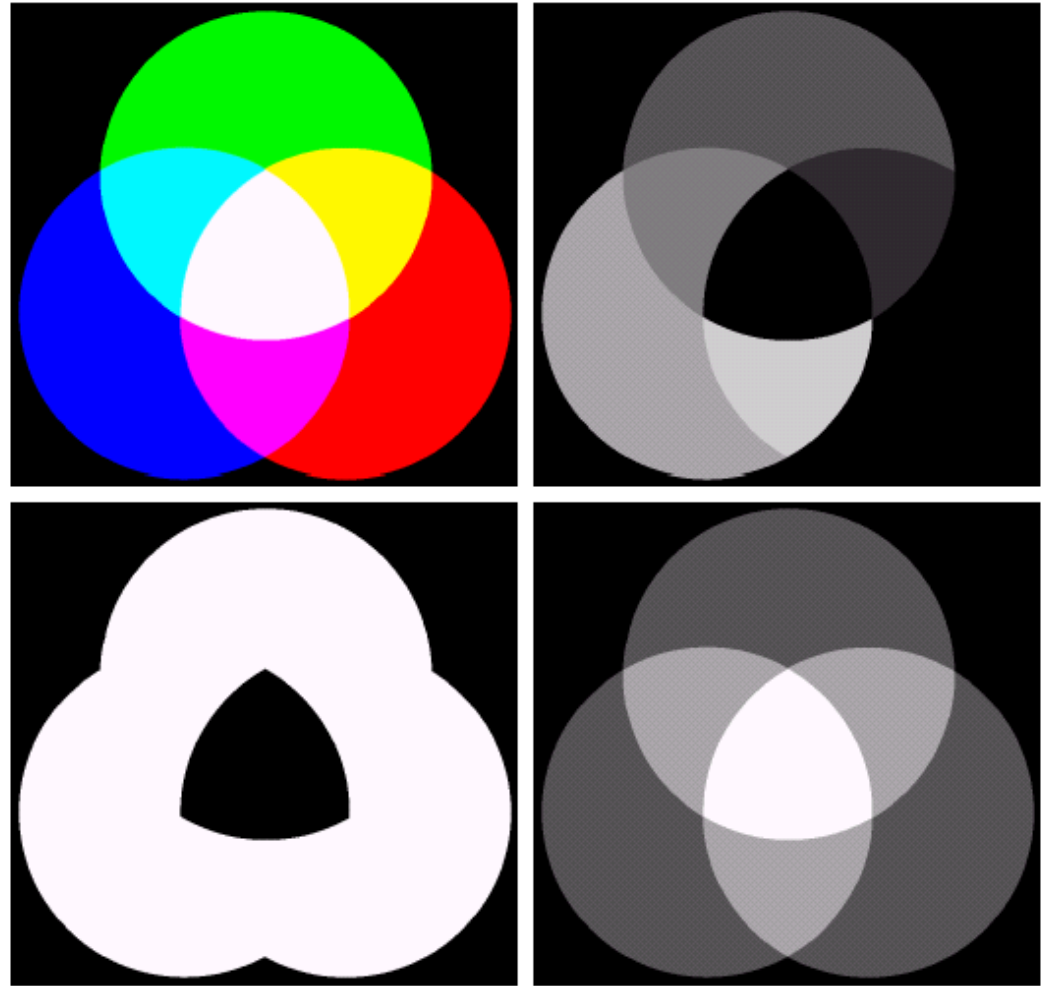
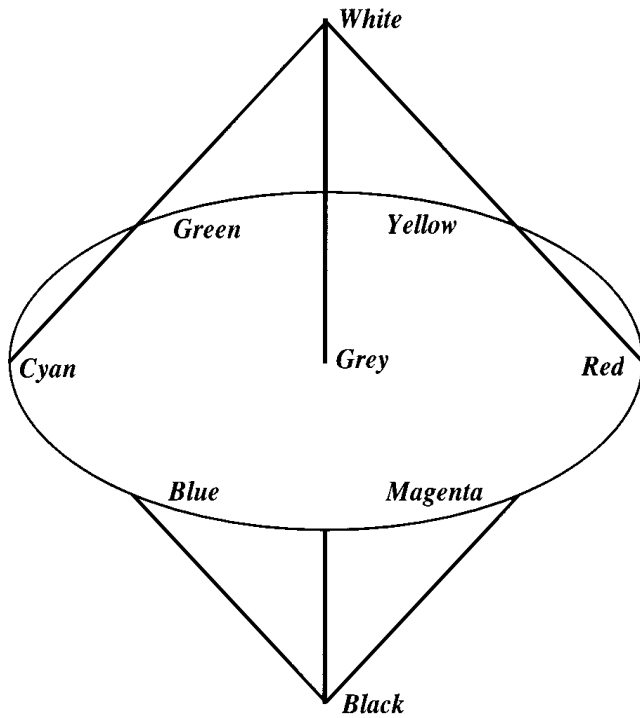
$$B = 1 - (R + G)$$

BR sector ($240 \leq H < 360$)

$$G = I(1 - S)$$

$$B = I \left[1 + \frac{S \cos(H - 240)}{\cos(60 - (H - 240))} \right]$$

$$R = 1 - (G + B)$$



a	b
c	d

FIGURE 6.16 (a) RGB image and the components of its corresponding HSI image: (b) hue, (c) saturation, and (d) intensity.

COLOR IMAGE - HSI

H-Channel



S-Channel



I-Channel

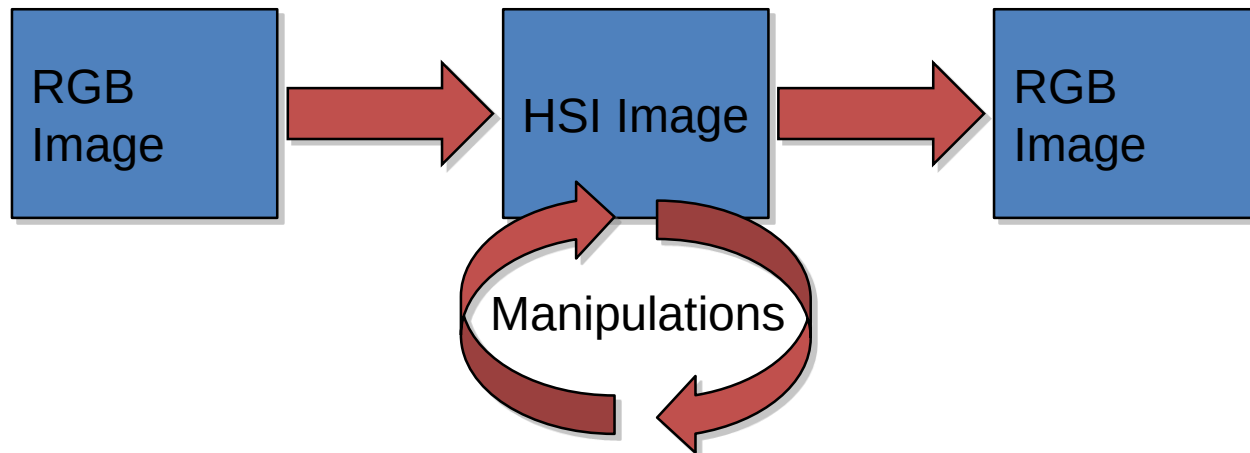


a b c

FIGURE 6.39 HSI components of the RGB color image in Fig. 6.38(a). (a) Hue. (b) Saturation. (c) Intensity.

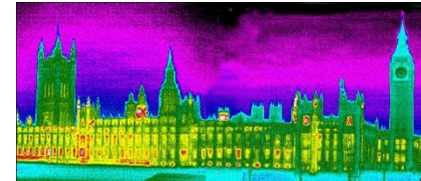
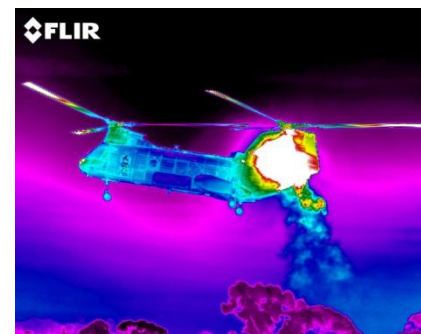
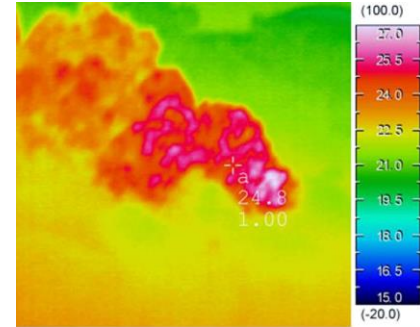
Manipulating Images In The HSI Model

- In order to manipulate an image under the HSI model we:
 - First convert it from RGB to HSI
 - Perform our manipulations under HSI
 - Finally convert the image back from HSI to RGB



Pseudocolor Image Processing

- Pseudocolor (also called false color) image processing consists of assigning colors to grey values based on a specific criterion
- The principle use of pseudocolor image processing is for human visualization



Pseudo Color Image Processing – Intensity Slicing

- Intensity slicing and color coding is one of the simplest kinds of pseudocolor image processing
- First we consider an image as a 3D function mapping spatial coordinates to intensities (that we can consider heights)
- Now consider placing planes at certain levels parallel to the coordinate plane
- If a value is one side of such a plane it is rendered in one color, and a different color if on the other side

Pseudo Color Image Processing – Intensity Slicing

- In general intensity slicing can be summarized as:
 - Let $[0, L-1]$ represent the grey scale
 - Let I_0 represent black [$f(x, y) = 0$] and let I_{L-1} represent white [$f(x, y) = L-1$]
 - Suppose P planes perpendicular to the intensity axis are defined at levels $I_1, I_2, \dots, I_P$
 - Assuming that $0 < P < L-1$ then the P planes partition the grey scale into $P + 1$ intervals $V_1, V_2, \dots, V_{P+1}$

Pseudo Color Image Processing – Intensity Slicing

- Grey level color assignments can then be made according to the relation:

$$f(x,y) = c_k \quad \text{if } f(x,y) \in V_k$$

- where c_k is the color associated with the k^{th} intensity level V_k defined by the partitioning planes at $l = k - 1$ and $l = k$

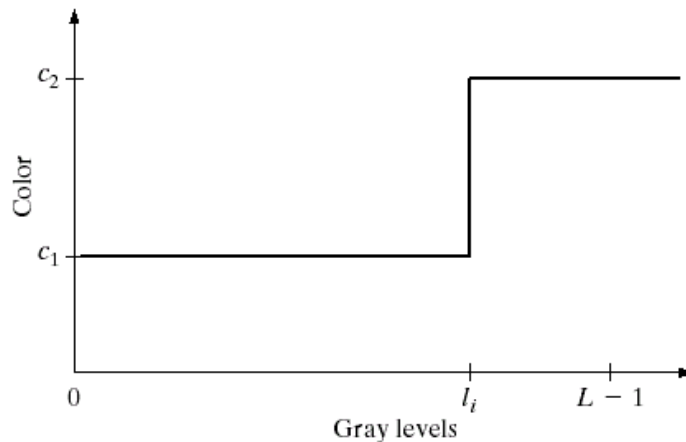
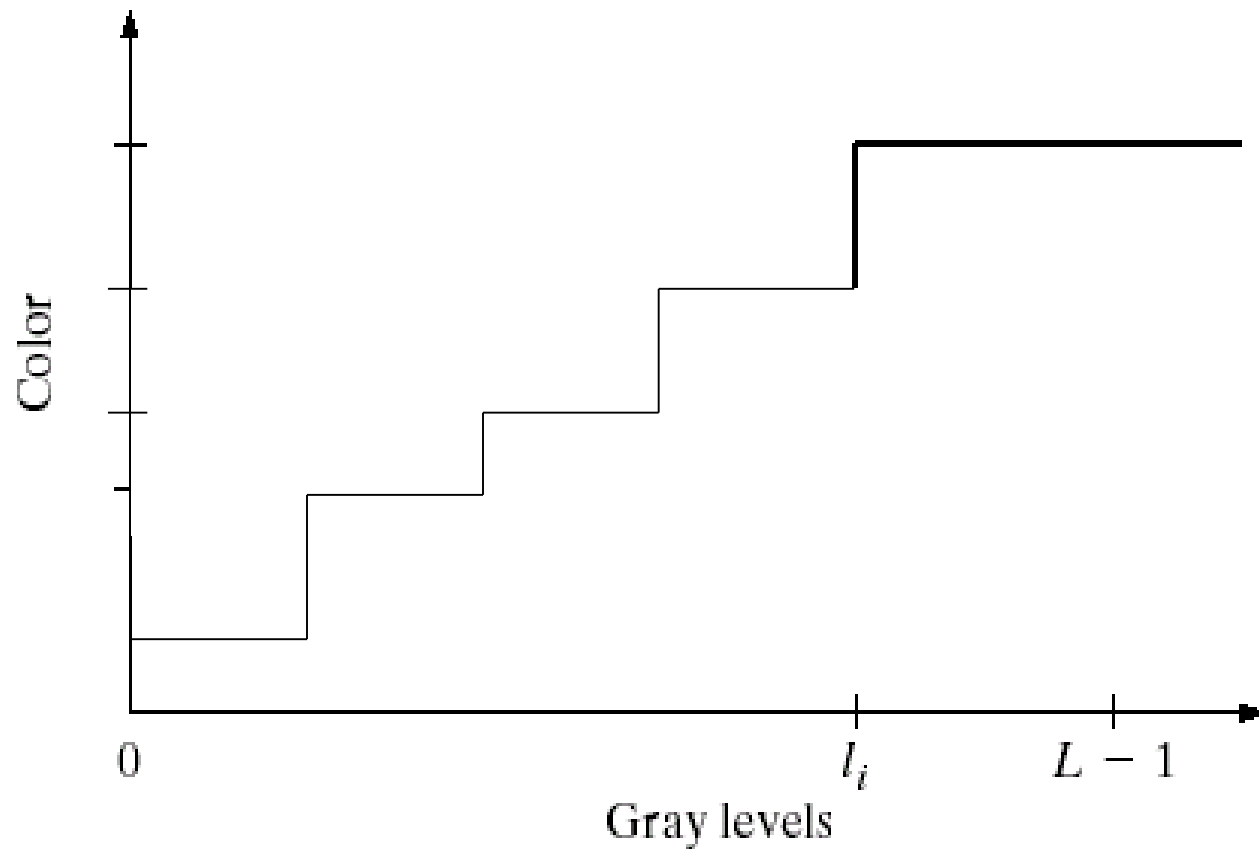


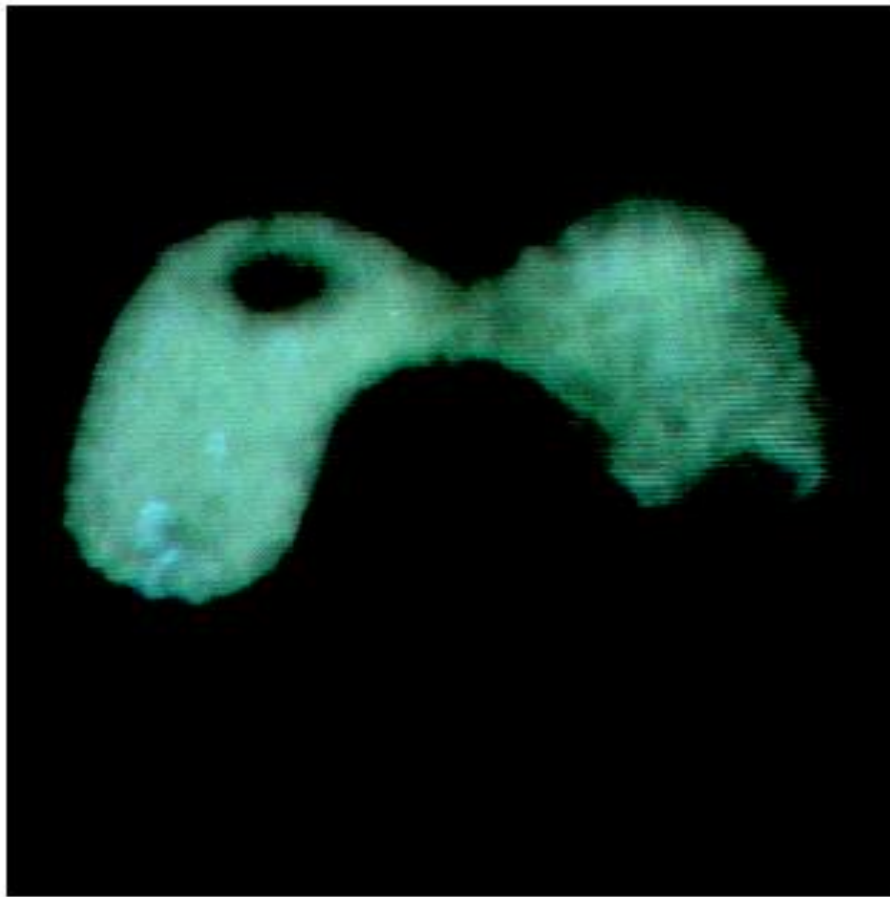
FIGURE 6.19 An alternative representation of the intensity-slicing technique.

Intensity slicing (cont.)

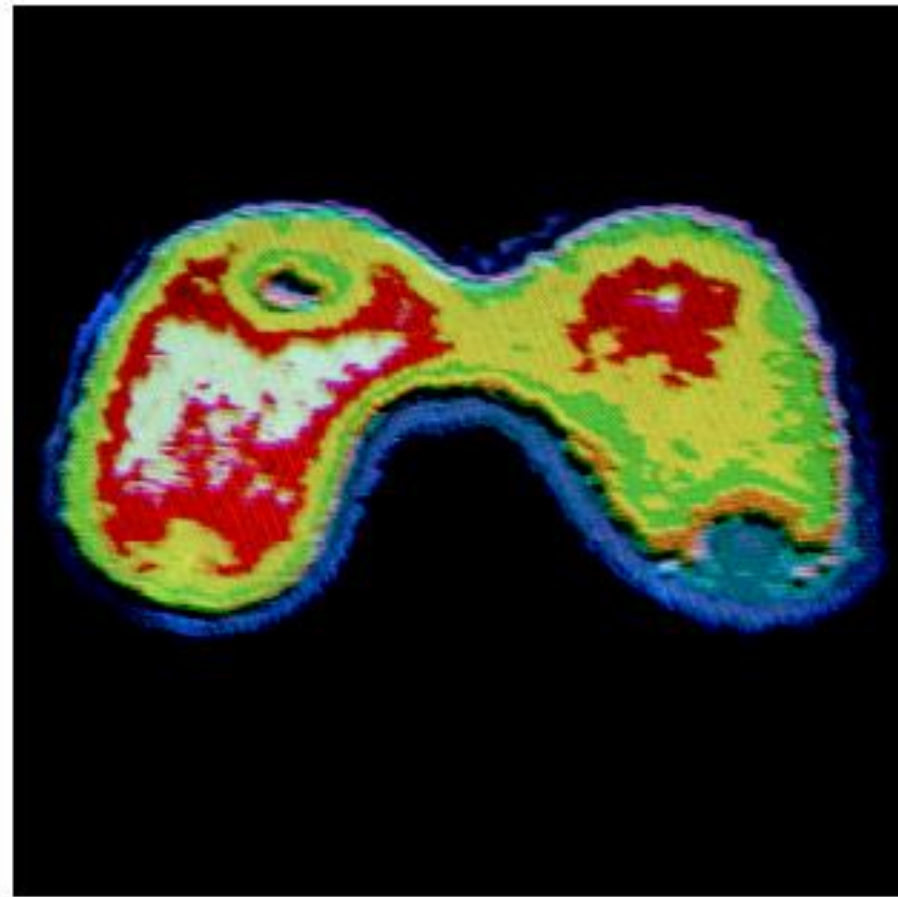
- More slicing plane, more colors



Pseudo Color Image Processing – Intensity Slicing



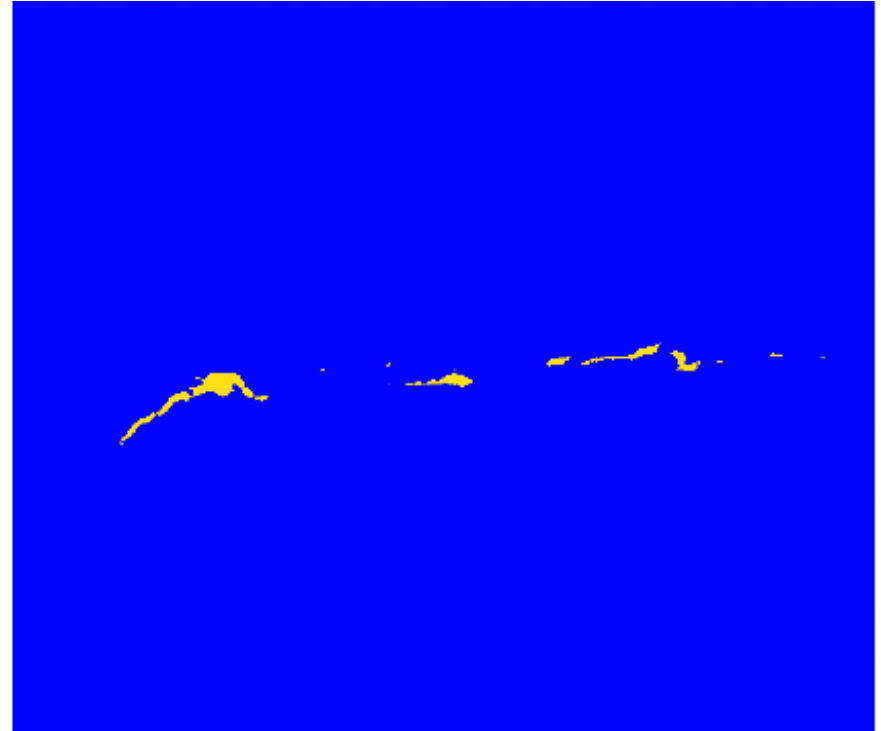
Radiation test pattern



8 color regions



Pseudo Color Image Processing – Intensity Slicing



Gray level to color transformation

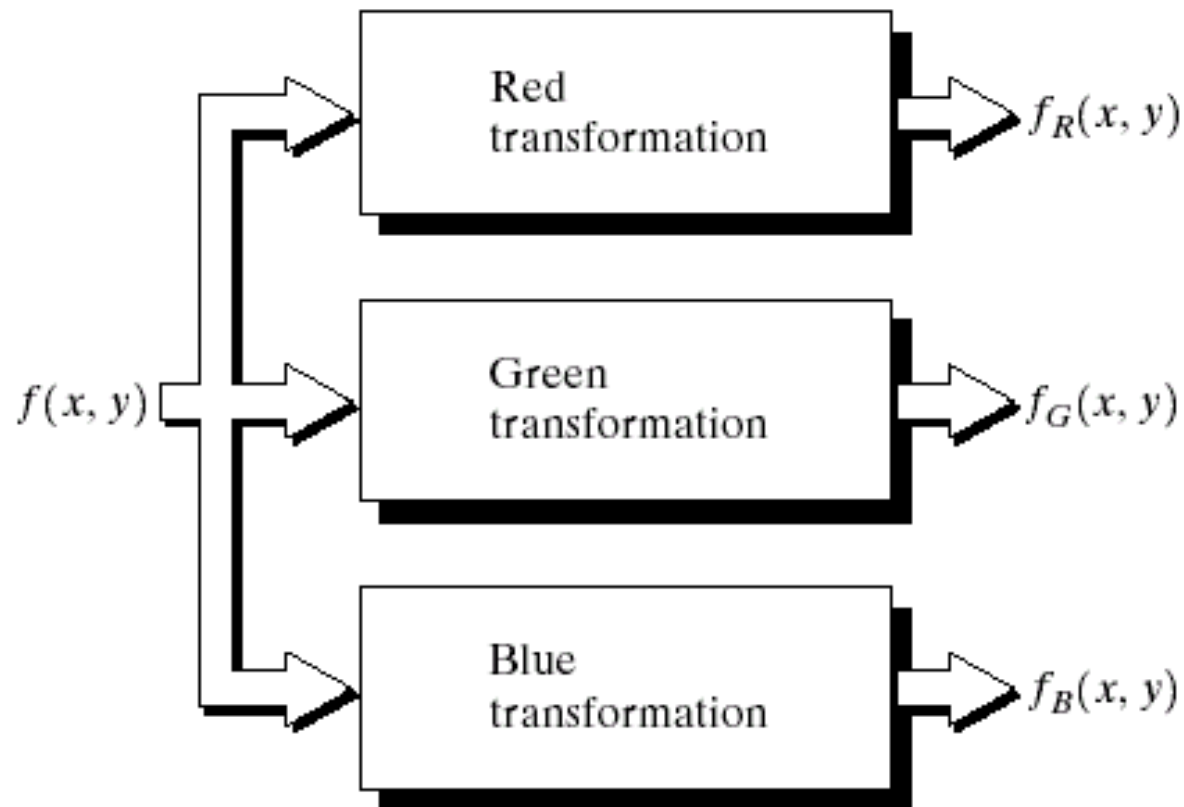


FIGURE 6.23 Functional block diagram for pseudocolor image processing. f_R , f_G , and f_B are fed into the corresponding red, green, and blue inputs of an RGB color monitor.

Color pixel

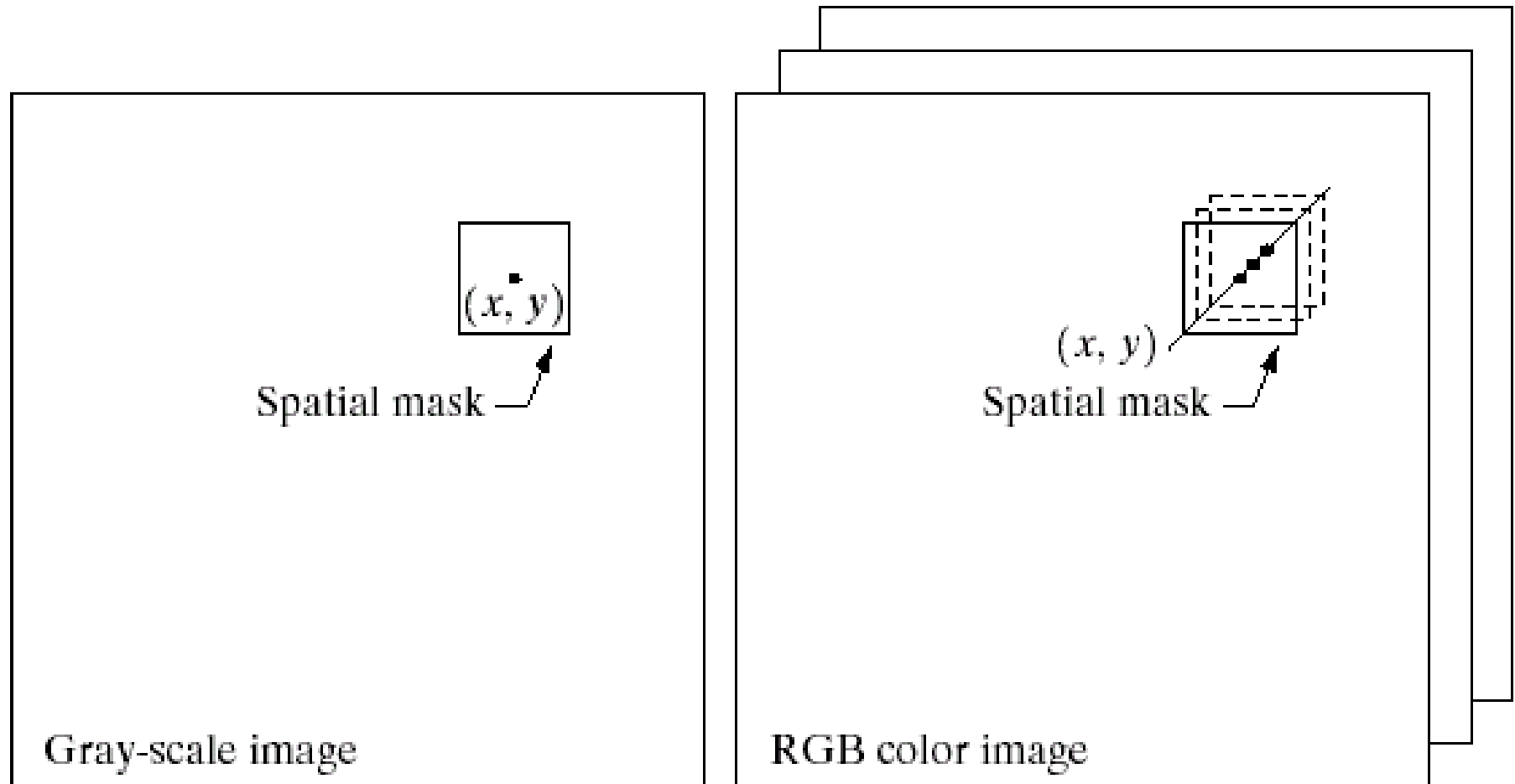
- A pixel at (x,y) is a **vector** in the color space
 - RGB color space

$$\mathbf{c}(x, y) = \begin{bmatrix} R(x, y) \\ G(x, y) \\ B(x, y) \end{bmatrix}$$

c.f. gray-scale image

$$f(x,y) = I(x,y)$$

Example: spatial mask



How to deal with color vector?

- 
- Per-color-component processing
 - Process each color component
 - Vector-based processing
 - Process the color vector of each pixel
 - When can the above methods be equivalent?
 - Process can be applied to both scalars and vectors
 - Operation on each component of a vector must be independent of the other component

Two spatial processing categories

- Similar to gray scale processing studied before, we have two major categories
- Pixel-wise processing
- Neighborhood processing

COLOR IMAGE - SMOOTHING

- Smoothing can be viewed as a spatial filtering operation in which the coefficients of the filtering mask are all 1's
- This concept can be easily extended to the processing of full-color images
- Simply smooth each of the RGB color planes and then combine the processed planes to form a smoothed full-color result

$$\hat{\mathbf{C}}(x, y) = \frac{1}{MN} \begin{bmatrix} \sum_{(x,y) \in S_{xy}} R(x, y) \\ \sum_{(x,y) \in S_{xy}} G(x, y) \\ \sum_{(x,y) \in S_{xy}} B(x, y) \end{bmatrix}$$

Color Image Sharpening

The Laplacian of vector c is

$$\nabla^2 [c(x, y)] = \begin{bmatrix} \nabla^2 R(x, y) \\ \nabla^2 G(x, y) \\ \nabla^2 B(x, y) \end{bmatrix}$$

Color Edge Detection (1)

Let r , g , and b be unit vectors along the R, G, and B axis of RGB color space, and define vectors

$$\mathbf{u} = \frac{\partial R}{\partial x} \mathbf{r} + \frac{\partial G}{\partial x} \mathbf{g} + \frac{\partial B}{\partial x} \mathbf{b}$$

and

$$\mathbf{v} = \frac{\partial R}{\partial y} \mathbf{r} + \frac{\partial G}{\partial y} \mathbf{g} + \frac{\partial B}{\partial y} \mathbf{b}$$

Color Edge Detection (2)

$$g_{xx} = \mathbf{u} \square \mathbf{u} = \left| \frac{\partial R}{\partial x} \right|^2 + \left| \frac{\partial G}{\partial x} \right|^2 + \left| \frac{\partial B}{\partial x} \right|^2$$

$$g_{yy} = \mathbf{v} \square \mathbf{v} = \left| \frac{\partial R}{\partial y} \right|^2 + \left| \frac{\partial G}{\partial y} \right|^2 + \left| \frac{\partial B}{\partial y} \right|^2$$

and

$$g_{xy} = \mathbf{u} \square \mathbf{v} = \frac{\partial R}{\partial x} \frac{\partial R}{\partial y} + \frac{\partial G}{\partial x} \frac{\partial G}{\partial y} + \frac{\partial B}{\partial x} \frac{\partial B}{\partial y}$$

Readings from Book (3rd Edn.)

- Color Processing Chapter-6



Acknowledgements

- ◆ Digital Image Processing”, Rafael C. Gonzalez & Richard E. Woods, Addison-Wesley, 2002
- ◆ Computer Vision: Algorithms and Applications Richard Szeliski